A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 20

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1 Stress-related illness:** ulcers, high blood pressure, heart disease **1 Behavioural changes:** irritability, aggression, lack of concentration leading to accidents **1** Communication problems / educational underperformance (eg schools under flight paths) **1** Sleep disturbance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 3 points below. **1 Acoustic fatigue:** objects exposed to sound at their natural resonant frequency vibrate, causing stress cracking / structural failure (eg early jet engine cowlings) **1 Vibration damage:** repetitive vibration from heavy vehicles damages buildings, bridges, underground pipes **1 Shock impacts:** the force of a sudden very loud noise (sonic boom) breaks windows / dislodges masonry **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Birds that sing to establish territories / attract mates cannot be heard against background noise **1** Birds scared by sudden noise leave nests so predators take eggs / chicks – cliff-nesting seabirds knock eggs or chicks into the sea **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** Whales / dolphins use hearing / echolocation to find food and communicate – loud sonar causes hearing damage if close to the source **1** Behavioural changes / disorientation possibly linked to mass strandings – reduced feeding success **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 287; AQA spec 3.4.3.2.13.	
02.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - Steam washing (hot water / steam jets kill organisms in the sediment)	1
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Chemicals cause the oil molecules to join together / solidify **1** Producing more solid material that can be collected / lifted more easily than liquid oil **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Air contains about 210 cm^3 of oxygen per litre whereas water at 20 °C holds a maximum of about 6 cm^3 per litre (less than 3% of the amount in air) **1** So a small amount of oxygen-consuming activity (decomposition, respiration) or warming can remove most of the oxygen – many aquatic organisms are near the lower end of their range of tolerance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **deoxygenation** → loss of dissolved oxygen from the water • **range of tolerance** → the range of conditions in which a species can survive	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 254-262; AQA spec 3.4.3.2.6-3.4.3.2.7.	
03.1	**1 mark each** 1 mark for each of the 4 points below. **1** CO2 rises from about 412 ppm in January to a peak of 416 ppm in May, falls to a minimum of 407 ppm in September, then rises again **1** During spring / summer, photosynthesis by plants (growing season) removes CO2 faster than respiration releases it, so the concentration falls **1** In autumn / winter photosynthesis is reduced (leaf fall) but respiration and decomposition continue, so CO2 rises **1** The atmosphere is in dynamic equilibrium: the concentration fluctuates around a mean – more land / vegetation in the northern hemisphere makes the cycle large **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Natural processes that add gases (respiration, combustion, volcanoes, denitrification) are balanced by processes that remove them (photosynthesis, dissolving, nitrogen fixation) **1** So the average composition stays the same over long timescales although the rates vary and concentrations fluctuate around a mean **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **denitrification** → nitrate converted back to nitrogen gas • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds	2

Do not write outside the box

Question	Marking guidance	Marks
03.3	**Any 2 of 5** 1 mark each for any 2 of the 5 points below, in any order, maximum 2. **1** Absorbs harmful UV / solar wind radiation in the stratosphere (ozone layer) **1** Greenhouse gases delay the escape of infrared, keeping the surface warm (33 °C warmer than without) **1** Atmospheric pressure allows liquid water to exist **1** Distributes heat from the tropics to higher latitudes by winds **1** Transports water vapour to areas that would otherwise be dry **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ozone layer** → the ozone in the stratosphere that absorbs UV • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	2
03.4	**1 mark each** 1 mark for each of the 2 points below. **1** Troposphere **1** Stratosphere **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-89; AQA spec 3.2.1.1.	
04.1	**1 mark each** 1 mark for each of the 3 points below. **1 Renewable:** re-forms naturally relatively quickly so use does not necessarily reduce future availability (solar, wind, tidal, geothermal, biofuel) **1 Depletable:** use can reduce future availability - all non-renewables (fossil fuels, uranium) plus renewables exploited unsustainably **1** Wood / biofuels are renewable but depletable if forests are felled faster than they regrow **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	3

Do not write outside the box

Question	Marking guidance	Marks
04.2	**1 mark each** 1 mark for each of the 3 points below. **1** Other factors (technology, location, cost) restrict how much can be harnessed **1 Examples:** coal too deep to mine – winds high above the ground / far offshore where turbines cannot be placed – waves in deep water where equipment cannot be anchored – low-intensity sunlight that cannot produce high temperatures **1** New technologies or higher prices may make these usable in future **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
04.3	**Any 4 of 7** 1 mark each for any 4 of the 7 points below, in any order, maximum 4. **1 Wind:** strong reliable winds - shallow seas, open plains, uplands – low land-use conflict (not near urban areas, ecologically sensitive or scenic areas) – near the grid **1 HEP:** high reliable rainfall, large catchment, narrow valley for a small dam with a large basin, impermeable stable bedrock **1 Solar:** high light intensity, low cloud cover **1 Wave:** coasts with strong reliable prevailing winds and a long fetch **1 Tidal:** large tidal range, coastal features (estuaries, channels) that focus the flow **1 Geothermal:** hot rocks near the surface / recent volcanic activity **1 Biofuels:** nearby farmland / forest / urban waste to avoid transporting bulky low-density fuel **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated • **HEP** → electricity generated from falling water • **impermeable** → does not let water through **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-177; AQA spec 3.3.2.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**1 mark each** 1 mark for each of the 2 points below. **1** Photons of light absorbed by the (n-type silicon) upper layer dislodge electrons, making it negatively charged **1** Electrons flow through an external conductor / circuit to the relatively positive lower layer – the current powers appliances **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'solar panels do not work in cold countries' - output depends on insolation, and cells are slightly more efficient when cool	2
05.2	**1 mark each** 1 mark for each of the 3 points below. **1** Smooth cells reflect about 30% of the light – grooved / textured surfaces (mimicking moth eyes) reflect light into the cell rather than away **1** Multi-junction cells have several layers of different materials, each absorbing different wavelengths **1** So a greater proportion of the available light is absorbed and converted to electricity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
05.3	**1 mark each** 1 mark for each of the 3 points below. **1 Solar farm:** polycrystalline / thin film - lowest cost per unit electricity where land area is not limiting (efficiency per m^2 matters less) **1 Satellite:** multi-junction - highest efficiency where mass / area is critical and cost is less important **1** Cost per watt = cost / efficiency: thin film 0.125, poly 0.125, mono 0.136, multi-junction 0.625 per unit **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
05.4	**1 mark each** 1 mark for each of the 2 points below. **1 Intermittent:** no output at night and low in winter (daily and seasonal cycles) - predictable **1 Unpredictable:** cloud cover, dust and smoke reduce intensity and scatter light and cannot be forecast accurately **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 199-203; AQA spec 3.3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**1 mark each** 1 mark for each of the 2 points below. **1** Numbers fell from about 210 pairs in 1945 to a minimum of about 25 in 1965 (steepest fall 1955-1960) **1** Numbers then recovered / rose steadily after 1970, reaching about 190 by 1985 (not yet back to 1945 level) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (200 - 25) / 200 x 100 **1** = 87.5% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
06.3	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** DDT is persistent (remains for decades) and liposoluble so it bioaccumulates in prey (small birds) and biomagnifies in sparrowhawks (top predator) **1** Sub-lethal doses caused thin eggshells / reduced breeding success so the population declined even without adult deaths **1** Higher doses caused sterility / death **1** After restrictions, environmental concentrations slowly fell (persistence explains the delay before recovery) **1** Breeding success recovered so the population rose **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistent** → does not break down / stays in the environment for a long time	4
06.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** DDT persists in soil / sediments / food chains for many years so exposure continued **1 Birds of prey are k-selected:** few young per year, slow to mature so populations grow slowly **1** Other pressures (habitat loss, persecution) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-266; AQA spec 3.4.3.2.8.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**1 mark each** 1 mark for each of the 3 points below. **1** 2020 has higher peak flow in the wet season (32 vs 22 m^3 s^{-1} in March) **1** 2020 has much lower dry-season flow (2 vs 9 m^3 s^{-1} in July / August) **1** Flow is much more variable / seasonal in 2020 – the river almost dries up **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	3
07.2	**1 mark each** 1 mark for each of the 3 points below. **1** In 1990 snow / ice stored wet-season precipitation and released it gradually as meltwater during the dry season, evening out the flow **1** With less ice, precipitation now falls as rain / melts immediately and runs off quickly, increasing wet-season flow and flood risk **1** Little stored ice remains to melt in the dry season so flow falls very low **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	3
07.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Water shortages for irrigation / drinking / HEP in the dry season **1** Flooding in the wet season **1** Sudden floods from ice-lake outbursts (glacial lake outburst floods) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **HEP** → electricity generated from falling water **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
07.4	**1 mark each** 1 mark for each of the 2 points below. **1** Surface meltwater flows down cracks to the base, lubricating the ice so it slides faster – if this exceeds the faster melting at the front, the front advances **1** But if snowfall at the source has not increased, total volume falls – both retreating and extending glaciers can indicate warming **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 94-96; AQA spec 3.2.1.2.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**1 mark each** 1 mark for each of the 3 points below. **1** t-test **1** The data are measured (continuous) values and two means are being compared **1** The data are (assumed to be) normally distributed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **t-test** → a test for a difference between the means of two normally distributed sets **Do not credit** • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	3
08.2	1 mark for the point below. **1** Use the Mann-Whitney U test instead (compares medians) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Mann-Whitney** → a test for a difference between two sets of unpaired, non-normal data **Do not credit** • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	1
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** There is a 2% probability that the difference occurred by random chance **1** As this is less than 0.05 the difference is statistically significant / the null hypothesis is rejected **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **random** → positions chosen using random numbers so the choice is not biased • **statistically significant** → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance • **null hypothesis** → a statement that there is no significant difference or correlation **Do not credit** • 'the hypothesis was proved' - a statistical test either rejects the null hypothesis or fails to reject it	2
08.4	**1 mark each** 1 mark for each of the 2 points below. **1** The ranges of values (mean plus or minus SD) do not overlap: woodland up to 1.6 and clearing down to 2.4 **1** About 68% of values lie within one SD of each mean so the two data sets are clearly separated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
08.5	**1 mark each** 1 mark for each of the 2 points below. **1** Significance shows the difference is unlikely to be due to chance, not that a causal relationship exists **1** Another variable (eg topography / aspect of the sites) could cause both – a controlled experiment or further evidence is needed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **statistically significant** → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance • **causal** → one variable actually causing the change in the other	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 420-423; AQA spec 3.7.2.3.	
09.1	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Volume of ash is much less than the original waste (less landfill) **1** Heat used for district heating / electricity generation (energy recovery) **1** No sorting / complicated management needed **1** Destroys pathogens / flammable organic wastes converted to CO2 and water **1** Refuse-derived fuel can replace fossil fuels (eg cement kilns) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	3
09.2	**1 mark each** 1 mark for each of the 4 points below. **1** High temperature and rapid cooling reduce the formation of toxic dioxins (from organic wastes reacting with chlorine) **1** Scrubber dissolves hydrogen chloride (from chlorine in PVC) which is then neutralised with alkali / limestone **OR** removes heavy metals for storage at high pH **1** Wet waste needs extra fuel to burn and lowers combustion temperature – composting recovers its nutrient value **1** Recycling conserves the resource value of metals, glass and plastics that would otherwise be lost **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Recycling** → reprocessing a material to make something new **Do not credit** • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4

Do not write outside the box

Question	Marking guidance	Marks
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** Ash may contain concentrated heavy metals / toxic residues **1** Must be stored in solid form at high pH / hazardous waste landfill so the metals remain insoluble **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
09.4	**Any 1 of 2** 1 mark each for any 1 of the 2 points below, in any order, maximum 1. **1** Fuel (natural gas) needed to burn wet / non-flammable wastes **1** Cost of treating effluent gases (SO2, NOx, HCl, smoke) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **incineration** → burning waste, usually with energy recovery	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 283-284; AQA spec 3.4.3.2.12.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **State of matter:** solids deposit near the source, gases disperse widely in the atmosphere – density (lead dust, hydrogen cyanide at Bhopal) controls how far particles/gases travel • **Persistence / half-life:** persistent pollutants (organochlorines, CFCs, heavy metals, radioactive wastes with long half-lives) travel further, accumulate and need containment rather than dilution – non-persistent pollutants (sewage, organophosphates) can be allowed to degrade • Toxicity, specificity, mode of action (enzyme inhibition, neurotoxins) determine acceptable concentrations and which species are at risk – LD50 / MDAF / MDNF used to set doses • Solubility in water (nitrates disperse and leach - control by nitrate vulnerable zones) vs liposolubility (DDT, mercury bioaccumulate and biomagnify - control by bans / restrictions) • **Reactivity:** reactive pollutants may degrade quickly or form secondary pollutants (ozone, acid rain, PANs) - controlling primary pollutants (catalytic converters, FGD) prevents secondary pollutants • Adsorption onto clay / sediments immobilises metals and caesium – can be used deliberately (activated carbon filters, polymer adsorption) • Mutagenic / carcinogenic / teratogenic action means no safe dose – exposure kept As Low As Reasonably Achievable (ALARA) • Synergism (ozone + SO2; smoke + SO2) means pollutants must be considered together • Environmental features (temperature inversions, currents, pH, oxygen) interact with properties: Critical Pathway Analysis uses properties + environment to predict where pollutants go and to focus monitoring • **Evaluation:** properties of new chemicals can be predicted from similar ones (precautionary principle)	9

Do not write outside the box

Question	Marking guidance	Marks
	– understanding allows prevention rather than clean-up **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **Persistence** → does not break down / stays in the environment for a long time • **half-life** → the time for the activity, or the amount, to fall by half • **CFCs** → CFCs / halogenated organic compounds releasing chlorine **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen	
	Model answer - Level 3 (9 marks)	
	A pollutant's properties largely determine where it goes and how much harm it does, so they can be used both to predict its behaviour and to choose a control. State of matter and density govern dispersal: solids deposit near the source, gases spread widely, and dense gases such as hydrogen cyanide at Bhopal hug the ground. Persistence decides whether dilution is enough: non-persistent wastes like sewage can be discharged where they break down, but organochlorines, CFCs, heavy metals and long-lived radioisotopes travel far and accumulate, so they need containment, bans or restrictions instead. Toxicity, specificity and mode of action, quantified by LD50 and used to set acceptable doses, identify which species are at risk. Solubility explains why nitrate leaches and is controlled by nitrate vulnerable zones, while liposolubility explains why DDT and methyl mercury bioaccumulate and biomagnify along food chains. Reactive pollutants may degrade quickly or form secondary pollutants - ozone, acid rain, PANs - so controlling the primary emissions with catalytic converters or flue-gas desulfurisation prevents the secondary ones. Adsorption on clay immobilises metals and caesium and is exploited deliberately in activated-carbon and polymer filters. Mutagens and carcinogens have no safe dose, so exposure is kept As Low As Reasonably Achievable, and synergism between ozone and sulfur dioxide, or smoke and sulfur dioxide, means pollutants must be assessed together.	
	Properties interact with the environment - inversions, currents, pH, oxygen - which is why Critical Pathway Analysis combines both to predict where pollutants concentrate and to focus monitoring. The greatest value of this understanding is preventive: the properties of a new chemical can be predicted from similar ones under the precautionary principle, allowing control before release rather than clean-up afterwards.	
	Weaker answer - Level 1 (2 marks)	
	Pollutants have properties like toxicity, persistence and solubility. Toxic pollutants are dangerous. Persistent pollutants like DDT last a long time in the environment and build up in food chains. Soluble pollutants dissolve in water and get into rivers. Knowing these properties helps scientists to predict what pollutants will do and to control them, for example by banning DDT.	
	What makes the difference	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 1 defines three properties in isolation; Level 3 pairs every property with a prediction *and* a control strategy, which is what the question asks.	
	• Level 3 uses a wide range of accurate examples (Bhopal, nitrate zones, methyl mercury, PANs, caesium adsorption); Level 1 uses DDT twice.	
	• The final paragraph links properties to environmental conditions (CPA) and to the precautionary principle - synoptic evaluation that distinguishes the top band.	
10.2	1 mark for the point below. **1** The increase in the amount / concentration of a substance within an (individual) organism over time, because intake exceeds excretion or breakdown **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **bioaccumulation** → a pollutant building up inside one organism over time **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240; AQA spec 3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Scale of the problem:** how energy is used (transport, buildings, industry, water) – links between affluence and per capita use • **Transport:** bulk / integrated transport, ATM, vehicle design (aerodynamics, mass, tyres, KERS, combustion efficiency), low embodied energy, design for end of life, driver behaviour, electric vehicles • **Buildings:** orientation and passive solar, SA:V, thermal mass, low embodied energy materials, earth sheltering, glazing (low-e, inert gas, vacuum), insulation optimum, heat exchangers, automatic ventilation, occupancy sensors, thermostats, low-energy appliances, behaviour • **Industry:** heat recovery and exchangers, insulation and high-volume storage, CHP, integrated manufacture, recycling and mass reduction, high-voltage grid, peak shaving, ICT management • **Limits:** rebound effect, cost and payback, existing building stock, embodied energy of replacements, behaviour • Evaluation with quantitative examples – comparison with the cost of new supply	25
	– conclusion that conservation is the cheapest first step but cannot alone meet demand growth *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 170-174, 225-235; AQA spec 3.3.4.2, 3.3.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Composition and structure:** N_2, O_2, Ar, CO_2, ozone – troposphere and stratosphere – pressure and liquid water – gases for biological molecules and industry • **Energy processes:** UV absorbed by O / O_2 / O_3 dynamic equilibrium – visible light absorbed at the surface – infrared absorbed by greenhouse gases (33 °C warming) – heat distribution by winds and ocean currents – transport of water vapour – energy budget and albedo • **Global climate change:** anthropogenic greenhouse gases, sources, feedback and tipping points, consequences for climate, sea level, currents, species – difficulties of monitoring and prediction – control strategies and agreements • **Ozone depletion:** CFCs and halons, catalytic destruction, effects of UV, Montreal Protocol and replacements, recovery time • **Other threats:** aerosols, smoke, land-use change – interconnection of processes	25
	• **Evaluation:** which functions are most at risk – success of ozone controls vs slow progress on greenhouse gases – justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-113; AQA spec 3.2.1.	

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